CASE: Frequency-Domain Consistent Remote Sensing Image Synthesis With Context-Aware Dynamic Semantic Fusion

Junde Liu, Danpei Zhao, *Senior Member, IEEE*, Bo Yuan, Wentao Li, Tian Li, and Zhenwei Shi, *Senior Member, IEEE*

***Abstract*—Recent advances in large-scale pretrained text-to-image models have substantially enhanced semantic image synthesis. Nevertheless, generating high-fidelity remote sensing images (RSIs) remains challenging, particularly in preserving structural integrity and achieving precise semantic consistency across multiscale structures and complex scene contexts. In this article, we propose the Context-Aware dynamic SEmantic (CASE) fusion module, a novel framework for RSI synthesis that effectively unlocks the generative potential of pretrained models to address these challenges. Specifically, we construct a context-aware semantic map (CSM) to explicitly encode spatial layouts together with contextual information. Subsequently, a dynamic residual semantic fusion mechanism is formulated to adaptively integrate layout-level, contextual, and global textual semantics, enabling the synthesis of RSIs with coherent structures and realistic fine-grained details. In addition, a frequency constraint (FC) loss is proposed to regularize the frequency-domain characteristics of the synthesized images, further enhancing structural fidelity across scales. Extensive experiments demonstrate that CASE consistently improves structural integrity, semantic fidelity, and visual realism of synthesized RSIs. Moreover, when applied to data augmentation for downstream segmentation tasks, our method improves downstream segmentation performance under the evaluated augmentation protocol, boosting semantic segmentation performance by +2.89% d-mIoU on the challenging GID-15 benchmark.**

***Index Terms*—Diffusion model, discrete wavelet transform (DWT), image segmentation, remote sensing images (RSIs), semantic image synthesis.**

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I. Introduction

SEMANTIC image synthesis aims to generate high-quality and realistic images that are aligned with the given semantic masks. This form of conditional image synthesis plays a pivotal role in various applications, such as change detection [1], [2], [3], image segmentation [4], [5], [6], [7], [8], [9], and data enhancement [10], [11], [12], [13].

Early semantic image synthesis methods [14], [15], [16] relied primarily on generative adversarial network (GAN) [17] architecture, achieving notable results in specific domains. They often achieved good results by innovating semantic-mask embeddings and generation architectures. However, due to limitations in the scale of the training dataset and the representation of the semantic masks, the quality and diversity of the generated images often fall short. Later, the large-scale text-to-image generation models [18] demonstrated significant quality potential. Building on these pretrained text-to-image models (e.g., stable diffusion [19]), ControlNet [20], and T2I-Adapter [21] introduced an additional adapter to inject layout guidance but face issues with text semantics and layout fidelity. Recently, FreestyleNet [22] and PLACE [23] improved layout consistency by strengthening the connection between image tokens and text semantics, as shown in Fig. 1(b). However, these methods restrict image tokens to their respective semantic labels, overlooking the inherent correlations among different classes within the scene, which results in diminished semantic coherence and ultimately impedes the generation of high-fidelity images.

Since most existing methods are primarily designed for natural images or common items, applying them directly to remote sensing images (RSIs) often causes substantial performance degradation. Unlike natural scenes, remote sensing scenarios present several critical challenges to RSI synthesis.

1) *Semantic Ambiguity:* The mapping from semantic labels to textures is inherently ambiguous in RSIs. This is due to high intraclass variance and subtle interclass distinctions.
2) *Structural Coherence:* The RSIs demand fidelity at two conflicting scales: large-scale geographical continuity (e.g., coherent road) and fine-grained textures (e.g., repetitive patterns of crops).

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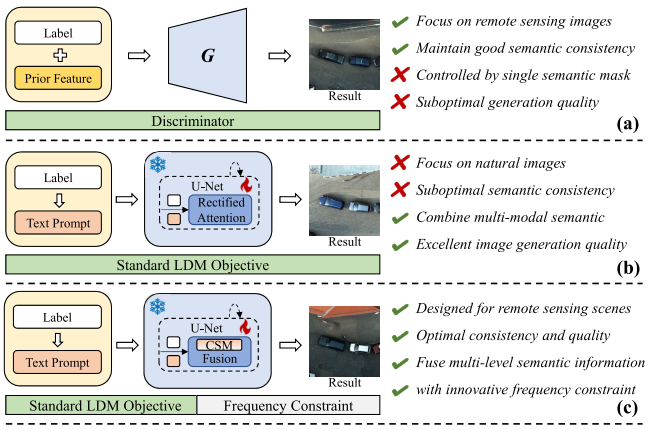


Fig. 1. Conceptual comparison motivating our proposed method. (a) GAN-based synthesis methods often struggle to generate high-quality, realistic textures. (b) Conversely, fine-tuning pretrained diffusion models can lead to poor semantic consistency, where the output deviates from the input layout. (c) Proposed CASE is engineered for remote sensing scenarios to strike a robust balance, achieving both high-fidelity synthesis and strict semantic adherence.

These traits frequently lead to semantic drift and discontinuities during generation. In recent years, CSEBGAN [7] and HySEGGAN [8] have generated semantic-controllable RSIs by introducing multidimensional discriminators and innovative embedding methods. However, due to the inherent limitations of GANs, the quality of the generated images remains suboptimal.

These advancements collectively underscore a bifurcated limitation in RSI synthesis. On the one hand, the GAN-based methods, despite incorporating specialized semantic embeddings, are fundamentally constrained by the architectural ceiling of the framework. This inherent limitation restricts the fidelity and textural diversity of the results. On the other hand, the approaches leveraging powerful pretrained diffusion models often fail to accommodate the unique geometric and frequency-domain characteristics inherent to RSIs. This leads to critical flaws in the synthesis, primarily manifesting as regional incoherence and pronounced semantic drift.

To address the aforementioned issues, we propose a Context-Aware dynamic SEmantic (CASE) fusion module, as shown in Fig. 2. The CASE leverages the pretrained stable diffusion model for high-quality, semantically coherent, and faithful RSI synthesis. The proposed method is specifically designed to transcend the quality limitations inherent in GANs while simultaneously anchoring the expressive power of pretrained stable diffusion models to the spatial context of remote sensing scenarios.

While foundational works have established effective layout-to-image control [7], [8], [24], they often overlook the complex semantic correlations inherent in remote sensing scenes. To address this gap, we present the context-aware semantic map (CSM), a novel mechanism designed to faithfully represent layout information by embedding it with context-aware features in the high-level feature space. Specifically, we explore the proportion of each text semantic component within the perceptual domain of each image token in the intermediate image features. Then, we use the context-aware semantic modulation module to generate the importance of the corresponding semantic components, resulting in a weighted proportion. Finally, we utilize the vector composed of these proportions as a CSM for the image token. While CSM accurately embeds layout and integrates text features with contextual semantics, we observe that solely relying on it can disrupt the mapping between image tokens and global text tokens, thereby affecting the visual quality of synthesized details. To effectively integrate CSM while preserving beneficial interactions for better visual quality, we develop a dynamic residual semantic fusion (DR Fusion) module. Particularly, for each fusion module, CASE learns an adaptive dynamic fusion parameter from the time embedding. This parameter is used to combine the CSM with the original cross-attention map. The resulting fusion map, which encapsulates both faithful layout information and multidimensional textual semantics, fundamentally enhances semantic coherence and the visual quality of the generated images. To address the unique frequency-domain characteristics and prevalent semantic drift inherent in RSI synthesis, we propose an effective frequency constraint (FC) loss to facilitate fine-tuning. The FC regulates the multiscale frequency-domain information of generated images, encouraging frequency-domain statistics closer to those of real images, enhancing visual realism and semantic coherence, as shown in Fig. 1(c).

The three components of CASE address complementary aspects of this problem: CSM models the spatial support of semantic tokens together with the contextual composition of the scene; DR Fusion preserves the original diffusion attention while adaptively injecting layout guidance across denoising timesteps; and FC loss regularizes the predicted clean latent in a multilevel directional wavelet representation to encourage agreement of coarse structures and oriented details.

The efficacy of our method is substantiated through extensive experiments on three challenging remote sensing datasets that encompass diverse geographical contexts. The results demonstrate that CASE achieves the best or competitive performance across most metrics. Although gains on some downstream segmentation metrics are modest, the consistent improvements in semantic consistency and data augmentation validate the effectiveness of the proposed design. Furthermore, to probe its robustness in data-limited conditions, we assessed its performance in a few-shot setting. These results demonstrate stability under the evaluated data-limited protocol rather than cross-domain or open-world generalization.

The contributions of this work are summarized as follows.

1) We propose a CASE fusion module for RSI synthesis, boosting the model's performance on downstream segmentation task with a frequency-context co-representation learning strategy.
2) We present a CSM to enforce precise layout fidelity, coupled with a dynamic residual fusion module that adaptively integrates multilevel semantic features, thereby ensuring robust structural integrity.
3) An effective FC loss is formulated to regularize the synthesized image, which preserves the rich textures and multiscale frequency details inherent in RSIs, mitigating semantic drift.

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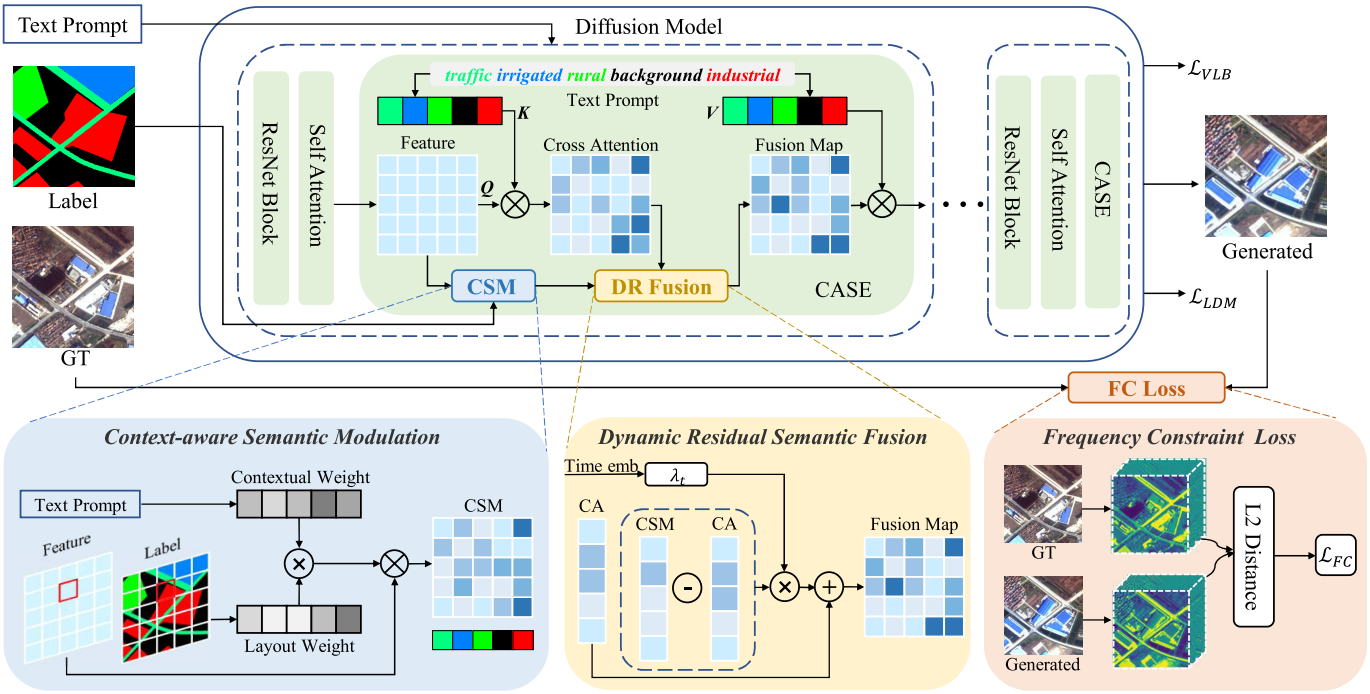


Fig. 2. Overview of our method. First, we leverage semantic masks and CASE-derived CSMs for semantic embedding, followed by feature fusion via a dynamic residual criterion. During fine-tuning, we combine $\mathcal{L}_{LDM}$, $\mathcal{L}_{VLB}$, and $\mathcal{L}_{FC}$ as the optimization objective.

4) Extensive evaluations on three challenging remote sensing datasets, GID-15, ISPRS Potsdam, and LoveDA Urban demonstrate the effectiveness of the proposed model and further validate its robustness under data-limited conditions.

II. Related Work

A. Semantic Image Synthesis

The semantic image synthesis aims to generate photorealistic images conditioned on semantic masks. Pioneering work Pix2Pix [14] framed this as an image-to-image translation task using conditional GANs (cGANs). To scale this approach to high resolutions, Pix2PixHD [15] introduced coarse-to-fine generators and multiscale discriminators, significantly improving realism.

A critical advancement was GauGAN [25], which introduced spatially adaptive normalization (SPADE). This layer modulates normalized activations using learned parameters derived from the semantic layout, effectively injecting spatial information throughout the generator. This architecture mitigates the semantic degradation often caused by standard normalization layers in deep networks. This spatially adaptive modulation became a dominant paradigm. The subsequent efforts [26], [27], [28], [29], [30] focused on refining it to achieve superior stylistic control and generative quality.

Beyond architectural refinement, the existing research tackled key challenges in fidelity and diversity. To improve semantic alignment and detail, the methods employed more sophisticated discriminators, such as segmentation-based objectives [31] or designed generators that explicitly model parameters for each semantic class independently [24]. Others focused on decomposing the generation process, using local class-specific and global image-level networks [32]. To address the one-to-many mapping problem (i.e., synthesis diversity), models introduced latent codes [33], utilized group convolutions [34], or parameterized the conditional convolution kernels [35], [36]. More recently, several approaches [37], [38], [39] have explored leveraging structural or shape information within the semantic mask to further enhance image quality and coherence.

Despite significant progress in semantic image synthesis, the existing methods still suffer from limited quality and diversity, primarily due to the small scale of the training dataset and the representation in the semantic mask.

B. Text-Based Semantic Image Synthesis

Text-to-image synthesis [29], [40], [41], [42] aims to generate realistic images based on a given global text prompt. Benefiting from the powerful diffusion model, the text-to-image synthesis demonstrates unprecedented generation quality and flexibility. In particular, the stable diffusion model [19], due to its robust stability and production quality, has become the most widely used foundational model in this field. The subsequent work has explored using pretrained models to achieve layout-controllable text-to-image synthesis [43], [44]. Ediffi [45] iteratively optimizes the alignment between the constrained cross-attention map and the target layout. However, they can only roughly control the position of synthesized objects. ControlNet [20] and T2I-Adapter [21] use an additional layout encoder to encode semantic maps but fail to ensure the constraint on layout consistency. Recently, FreestyleNet [22] introduced corrected cross-attention (RCA)

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to replace the cross-attention module in stable diffusion; however, the generated images lack sufficient layout alignment.

PLACE [23] improves semantic controllability and consistency through layout control maps. However, the layout control maps rely solely on the coarse-grained statistics of semantic masks and fail to integrate the contextual semantic information of tokens, resulting in insufficient local semantic coherence in the synthesized images. Moreover, most existing works primarily focus on the sufficient representation of the semantic layout, often neglecting the explicit modeling of contextual semantic dependencies. This omission leads to significant limitations in both visual quality and semantic coherence.

C. RSI Synthesis

In recent years, the semantic image synthesis has garnered widespread attention in remote sensing. However, due to the inherent complexities of RSIs, such as small interclass differences, large intraclass variations, and geographical continuity, directly applying advanced natural image synthesis methods often fails to achieve optimal results. Early GAN-based methods mainly focused on semantic modeling and fine-grained semantic refinement but still struggled with local semantic consistency and semantic coherence [7], [8]. CRS-Diff [46] has explored fine-tuning stable diffusion for text-to-image synthesis but still faces issues, such as semantic drift and suboptimal generation quality. More recently, the task-oriented synthesis methods [47], [48] have further explored the use of diffusion-based generation for downstream remote sensing semantic segmentation by introducing reference-semantic fusion and task-aware sampling strategies. However, these approaches primarily emphasize improving the utility of synthetic data for specific downstream tasks, rather than enhancing the semantic and structural fidelity of RSI synthesis through systematic fine-tuning of a pretrained diffusion model. Consequently, there remains a lack of effective diffusion-based frameworks that jointly address high-fidelity RSI synthesis and its benefits for downstream segmentation.

Furthermore, a significant disparity exists in the frequency-domain characteristics between RSIs and natural images, which further impedes the direct portability of conventional methods to remote sensing scenarios. Prior work [40], [49], [50], [51], [52] has largely overlooked this crucial domain gap in the frequency domain. To mitigate this, we integrate frequency-domain constraints into the generative process, providing an effective solution to preserve the structural integrity of RSIs.

III. Method

Semantic image synthesis aims to produce a photorealistic image I of height H and width W that is consistent with a given semantic mask $S \in \mathbb{R}^{H\times W\times C}$, where C denotes the number of semantic classes. Each pixel in S encodes a class label $k \in \{1, \ldots, C\}$, which specifies the semantic category expected at the corresponding spatial location in I. In the following, we detail the methodological components of CASE.

A. Preliminary: Stable Diffusion

Stable diffusion [19] is a foundational text-to-image generation model that operates as a latent diffusion model (LDM), which performs the computationally intensive diffusion process in a low-dimensional latent space. The U-Net denoiser ϵ_θ is trained to predict the noise ϵ that was added to the latent representation at a given timestep. The training objective for the conditional LDM is formulated as a simplified loss function that minimizes the mean squared error between the actual noise and the predicted noise. $\mathcal{L}_{LDM}$ can be computed as follows:

$$\mathcal{L}_{LDM} = \mathbb{E}_{\mathcal{E}(x),y,\epsilon\sim\mathcal{N}(0,1),t}\left[\|\epsilon - \epsilon_\theta(z_t, t, \tau_\theta(y))\|_2^2\right] \tag{1}$$

where y is the text prompt, t is a uniformly sampled timestep, and z_t is the noisy latent produced by adding noise ϵ to $z_0 = \mathcal{E}(x)$ at step t. ϵ_θ and $\tau_\theta(y)$ are the denoising network and the pretrained CLIP [53] text encoder, respectively.

In conditional LDM, text features are integrated into the intermediate layers of U-Net through a cross-attention module

$$\begin{aligned} Q &= W_Q \cdot \phi(z_t), \quad K = W_K \cdot \tau_\theta(y) \\ V &= W_V \cdot \tau_\theta(y), \\ \text{Attention}(Q, K, V) &= \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V \end{aligned} \tag{2}$$

where W_Q, W_K, and W_V are the learnable projection matrices of query, key, and value. $\phi(z_t)$ is the intermediate representation of the noisy latent within the U-Net.

The diffusion process can also be defined and optimized through the lens of a variational lower bound on the negative log-likelihood

$$\mathcal{L}_{VLB} = L_0 + L_1 + \cdots + L_{T-1} + L_T. \tag{3}$$

Each term L_t corresponds to a specific aspect of the diffusion and reverse processes. Specifically, L_0 is a reconstruction term, $L_{1\ldots T-1}$ are KL divergences regularizing the learned reverse transitions, and L_T is a prior term on the final latent z_T.

B. CASE Fusion Network

To address the challenges of semantic drift and discontinuity in RSI synthesis, we propose the CASE fusion module. Our method enhances a pretrained LDM by introducing a novel attention mechanism that precisely embeds layout information while preserving the rich semantic priors of the base model. As illustrated in Fig. 2, our approach consists of two core components: a context-aware semantic modulation module that generates a CSM, and a dynamic residual semantic fusion module that adaptively integrates this map with the original cross-attention mechanism.

1) Context-Aware Semantic Modulation: The standard cross-attention in diffusion models aligns image features with text semantics but lacks explicit spatial control, which is critical for generating RSIs with complex layouts. To enforce strict layout fidelity, we introduce a CSM that modulates the attention process based on both the desired layout and the global semantic context. Here, "context" refers to the global category context provided by the prompt and the local layout

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composition derived from the semantic mask. CASE is applied only to the cross-attention branches of the spatial transformer blocks at the three latent resolutions specified by the stable diffusion attention hierarchy. For a 512 × 512 input encoded into a 64 × 64 latent, these resolutions are 64 × 64, 32 × 32, and 16 × 16. The self-attention branches and nonattention ResBlocks remain unchanged.

Given an intermediate image feature representation $Z \in \mathbb{R}^{h\times w\times d_z}$ and a set of text token embeddings $E = [e_1; \ldots; e_N] \in \mathbb{R}^{N\times d_e}$, the cross-attention mechanism computes a similarity matrix between image queries Q and text keys K. Our goal is to modulate this similarity to be context-aware.

a) Semantic weight map: First, we compute a semantic weight map $W_s \in \mathbb{R}^{hw\times N}$. For each image patch i and text token j corresponding to a semantic class c_j, the weight $W_s(i, j)$ represents the proportion of pixels of class c_j within the spatial region of patch i. This is derived from the input semantic mask M, which is downsampled to match the feature map resolution

$$W_s(i, j) = \text{Pool}\left(M_{c_j}\right)_i \tag{4}$$

where M_{c_j} is the binary mask for class c_j and *Pool* is an average pooling operation. This map directly encodes the desired layout distribution of semantic classes.

However, the semantic meaning of a class in an RSI is often influenced by its neighbors. To capture this, we introduce a contextual weight ω_j for each text token e_j formulated as

$$\omega_j = H_c\left(e_j, \text{Aggregate}\left(E_{context}\right)\right). \tag{5}$$

This weight is dynamically generated by a small neural network, called the contextual weight head (H_c), which learns to interpret the importance of a token within the full text prompt. Here, $E_{context}$ represents the set of all valid token embeddings in the prompt, and *Aggregate* is a mean pooling operation, which allows the model to learn context-dependent textures.

b) Contextual weight head: To achieve a nuanced semantic modulation, we designed the contextual weight head module. Its primary function is to compute a dynamic, context-aware importance weight $\omega_j \in [0, 1]$ for each semantic token embedding e_j from the input prompt. This process is conditioned on both the specific token itself (as a query) and the holistic representation of the entire prompt (as global context). The weight generation process for a token e_j unfolds through the following stages.

First, the module separates the representation of the specific token from the global context. Both the query token e_j and the full context E are projected into a shared, lower dimensional latent space of dimension d_i using independent linear transformations

$$q_j^{proj} = W_q e_j + b_q \tag{6}$$

$$K^{proj} = EW_k + b_k \tag{7}$$

where $W_q, W_k \in \mathbb{R}^{d_e\times d_i}$ and $b_q, b_k \in \mathbb{R}^{d_i}$ are the learnable parameters. This yields a projected query vector $q_j^{proj} \in \mathbb{R}^{d_i}$ and a projected context matrix $K^{proj} \in \mathbb{R}^{N\times d_i}$.

To form a single vector representing the entire prompt's semantic meaning, we aggregate the projected context matrix K^{proj} using masked mean pooling. This ensures that padding tokens are ignored, providing a robust summary of the valid tokens. Given a binary mask $G_j \in \{0, 1\}^N$, where $G_j = 1$ for a valid token at position j, the aggregated global context vector $c_{global} \in \mathbb{R}^{d_i}$ is computed as

$$c_{global} = \frac{\sum_{j=1}^{N} G_j \cdot k_j^{proj}}{\sum_{j=1}^{N} G_j} \tag{8}$$

where k_j^{proj} is the jth row vector of the matrix K^{proj}.

The module then fuses the token-specific representation with the global context representation. This is achieved by concatenating the projected query vector q_j^{proj} and the aggregated context vector c_{global}

$$f_j = \left[q_j^{proj}; c_{global}\right]. \tag{9}$$

The resulting fused feature vector $f_j \in \mathbb{R}^{2d_i}$ now contains rich information about the token's identity as well as its relationship to the overall prompt.

Finally, the fused feature vector f_j is passed through a simple two-layer multilayer perceptron (MLP), which acts as the weight predictor. A GELU activation is used in the hidden layer, and a sigmoid function is applied to the output layer to ensure the final weight ω_j is normalized to the range $[0, 1]$. The process is formalized as

$$\omega_j = \sigma\left(\text{MLP}_\theta\left(f_j\right)\right) \tag{10}$$

where θ represents the learnable parameters of the MLP and σ is the sigmoid function. This final scalar ω_j is the predicted importance weight for the token e_j.

c) Context-aware semantic map: Finally, we generate the CSM by combining the semantic weight map and the tokenwise contextual guidance weight. Given the raw cross-attention logits $S_{ij} = (Q_i K_j^T)/\sqrt{d_k}$, we first keep the original cross-attention branch unchanged

$$A_{\text{orig}} = \text{softmax}(S). \tag{11}$$

In parallel, we build a mask-guided attention branch. For each text token j, the semantic weight $W_s(i, j)$ indicates the proportion of the corresponding semantic class within image patch i, and ω_j denotes the contextual guidance weight predicted from the token embedding and the aggregated valid prompt tokens. The mask-guided logits are formulated as

$$\tilde{S}_{ij} = \begin{cases} S_{ij} \cdot W_s(i, j) \cdot \omega_j, & W_s(i, j) > 0 \\ -\infty, & W_s(i, j) = 0. \end{cases} \tag{12}$$

The resulting CSM is computed as

$$A_{\text{CSM}} = \text{softmax}\left(\tilde{S}\right). \tag{13}$$

Here, W_s restricts each category token to its corresponding semantic regions, while ω_j adaptively calibrates the guidance strength according to the semantic composition of the prompt. Importantly, A_{CSM} is not used to replace the original cross-attention map directly. Instead, it serves as a layout-constrained attention prior that is later fused with A_{orig} through the dynamic residual semantic fusion module.

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2) Dynamic Residual Semantic Fusion: While A_{CSM} provides strong layout guidance, exclusively using it can disrupt the rich, pretrained knowledge of the base model, potentially degrading visual quality. To balance layout fidelity with generative quality, we propose a dynamic residual fusion mechanism. This mechanism adaptively interpolates between the original cross-attention map, $A_{orig} = softmax(S)$, and our CSM A_{CSM}.

We introduce a dynamic fusion parameter $\lambda_t \in [0, 1]$, which is predicted from the diffusion timestep embedding t_{emb} using a lightweight MLP with sigmoid activation, denoted as ϕ_{scale}

$$\lambda_t = \phi_{scale}(t_{emb}). \tag{14}$$

This enables the fusion strategy to adapt throughout the denoising process, emphasizing the strict layout guidance of A_{CSM} in early coarse generation stages (larger λ_t) and the fine-grained details and semantic coherence of A_{orig} during later refinement stages (smaller λ_t).

We adopt a residual fusion strategy to inject the condition-specific feature A_{CSM} into the backbone representation A_{orig}. The output explicitly preserves the identity mapping A_{orig} while learning only the corrective residual

$$A_{final} = A_{orig} + \lambda_t \left(A_{CSM} - A_{orig}\right). \tag{15}$$

The output of the attention layer is then calculated as $Output = A_{final}V$, where V is the value matrix.

C. Loss Function

1) FC Loss: To enhance the structural fidelity and textural realism of synthesized RSIs, we introduce a FC loss in the latent wavelet domain. In this setting, a real image x is first projected into a clean latent representation $z_0 = \mathcal{E}(x)$ by the encoder $\mathcal{E}$ of the pretrained first-stage autoencoder, and the diffusion U-Net is trained to predict the noise added to the latent variable. To maintain consistency with the LDM training pipeline, we instead formulate the proposed FC directly in the latent space. Specifically, given the noisy latent z_t at timestep t, the denoising network predicts the noise component $\epsilon_\theta(z_t, t, c)$ conditioned on the semantic and textual inputs c. Under the standard noise-prediction parameterization, the predicted clean latent $\hat{z}_0$ can be estimated as

$$\hat{z}_0 = \frac{z_t - \sqrt{1-\bar{\alpha}_t}\epsilon_\theta(z_t, t, c)}{\sqrt{\bar{\alpha}_t}} \tag{16}$$

where $\bar{\alpha}_t$ denotes the cumulative noise schedule. We then apply a 2-D discrete wavelet transform (DWT), denoted as $\mathcal{W}(\cdot)$, to both the predicted clean latent $\hat{z}_0$ and the ground-truth clean latent z_0. The transform decomposes the latent feature maps into one low-frequency component and multilevel directional high-frequency components

$$\begin{aligned}\left(\hat{z}_{LL}, \left\{\hat{z}_d^j\right\}_{d\in D, j=1}^{J}\right) &= \mathcal{W}(\hat{z}_0)\\ \left(z_{LL}, \left\{z_d^j\right\}_{d\in D, j=1}^{J}\right) &= \mathcal{W}(z_0)\end{aligned} \tag{17}$$

where $D = \{LH, HL, HH\}$ represents the horizontal, vertical, and diagonal high-frequency subbands, and J is the number of wavelet decomposition levels. The FC loss is defined as the sum of squared errors between the corresponding latent wavelet coefficients

$$\mathcal{L}_{FC}(\hat{z}_0, z_0) = \|\hat{z}_{LL} - z_{LL}\|_2^2 + \sum_{j=1}^{J}\sum_{d\in D}\left\|\hat{z}_d^j - z_d^j\right\|_2^2. \tag{18}$$

Although the proposed constraint is imposed in the latent domain rather than the RGB pixel domain, the latent representation produced still preserves spatial layouts and local textural patterns. The low-frequency component primarily constrains the coarse spatial structure and global layout, whereas the high-frequency components regularize discrepancies in local textures, fine-grained details, and directional structures, such as edges. Through the joint constraint across different frequency components, FC loss promotes the preservation of both global structural fidelity and local directional details in the generated latent representation, which are particularly important for RSIs. Accordingly, FC loss can be interpreted as a latent-domain, multiscale, direction-aware structural regularizer.

We implement the transform with the biorthogonal family, whose near linear phase and symmetric filters reduce boundary artifacts and preserve edges and fine details in RSIs. Section IV-E3 reports an ablation study over alternative wavelet bases.

2) Total Loss Function: We train the network end-to-end with a three-term loss: the standard LDM objective for semantic consistency, variational lower bound $\mathcal{L}_{VLB}$ for stable likelihood optimization, and FC loss $\mathcal{L}_{FC}$ to sharpen fine-grained details. The total loss function $\mathcal{L}_{total}$ is a weighted sum of these three components

$$\mathcal{L}_{total} = \lambda_{LDM}\mathcal{L}_{LDM} + \lambda_{VLB}\mathcal{L}_{VLB} + \lambda_{FC}\mathcal{L}_{FC} \tag{19}$$

where λ_{LDM}, λ_{VLB}, and λ_{FC} are the hyperparameters that balance the contribution of each term, with their default values set to 1, guiding the model to generate high-quality, realistic, and semantically coherent RSIs.

IV. Experiments

We compare our approach with several up-to-date semantic image synthesis methods, including Pix2PixHD [15], SPADE [25], CLADE [54], SAFM [28], INADE [24], Santa [55], CSEBGAN [7], HySEGGAN [8], FreestyleNet [22], PLACE [23], and CRS-Diff [46]. Among these, Pix2PixHD, SPADE, and SAFM are primarily quality-focused, while CLADE and INADE emphasize generating diversity. CSEBGAN and HySEGGAN are specialized for RSI synthesis. Santa represents a generation method for nonmatching data, while FreestyleNet, PLACE, and CRS-Diff are all based on fine-tuning stable diffusion. Bold and underlined fonts indicate the optimal and suboptimal values for each metric. Methods do not support certain types of evaluation, and their corresponding values are replaced with "-."

A. Experimental Settings

1) Experimental Details: Our model is implemented using PyTorch 2.2.1. All experiments are conducted on a Linux

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TABLE I
DETAILS OF THE DATASETS USED FOR THE EXPERIMENT

Dataset	Resolution	Type	Class	Train-num	Val-num
GID-15	0.8m	Satellite	16	1812	302
ISPRS	0.05m	Airborne	6	2712	462
LoveDA	0.3m	Satellite	7	1156	677

workstation equipped with a single NVIDIA GeForce RTX 4090 GPU with 24-GB memory. We utilize the pretrained V1-4 stable diffusion model [19] as the initialization weights and fine-tune it with a learning rate of $1e{-}6$. We fine-tuned for 80k iterations, and the batch size is 1. We employ 50 PLMS [56] sampling steps with a classifier-free guidance [57] scale of 2. For the FC loss, we employ a $J = 3$ level DWT using the biorthogonal wavelet. Symmetric padding is utilized to handle border effects during the decomposition. For the segmentation-based controllability evaluation, DeepLabV3+ is trained only on real training images and their corresponding semantic labels for 200 epochs. The model uses 512×512 inputs, batch size 16, mixed-precision training, and the Adam optimizer. The evaluator is frozen before being applied to synthesized images.

2) Dataset: The experiments were conducted across various remote sensing scenarios, including GID-15 [58], ISPRS [59], and LoveDA [60]. GID-15 contains 16 semantic categories. We used the fine land cover classification component, GID-15, which consists of 20 RSIs (6800×7200) along with their corresponding semantic masks. These images were divided into 256×256 image blocks, without overlap or intersection. The selected 2114 paired images were randomly split into training and validation sets at a 6:1 ratio, resulting in 1812 training images and 302 validation images.

The ISPRS is a classic aerial imagery dataset, of which Potsdam is used in this article. The ISPRS contains six semantic categories. For ISPRS Potsdam, we follow the same patch-based preprocessing protocol and use the prepared training and validation folders for all compared methods. It comprises 2712 training images and 462 validation images.

LoveDA is a large-scale remote sensing semantic segmentation dataset containing high-resolution aerial images collected from both urban and rural regions. In this article, we use only the Urban subset, which contains seven semantic categories covering typical urban land-cover objects. Following the official data partition, the urban subset comprises 1156 training images and 677 validation images.

During training, both the images and semantic maps are resized to 512×512. Since the input condition is a semantic mask rather than a free-form caption, we construct a class-name prompt for each sample by concatenating the class names that appear in the mask. The details of these datasets are shown in Table I. The model is trained solely on the available training set. For each original training image, we generate one corresponding synthetic image, yielding a synthetic-to-original ratio of 1:1. The semantic masks used by the image generator are taken directly from the dataset annotations to preserve semantic fidelity, doubling the downstream training set size.

3) Controllability Metrics: Following previous semantic image synthesis work [25], [28], the Inception-V3 [61] network is used in the FID [62] calculation to extract features from real and synthesized images. The last fully connected is removed to obtain a 2048-D pooled feature representation. Subsequently, the mean and covariance matrices are used to calculate the distance between the two high-dimensional distributions to compute the FID score as shown in the following equation:

$$FID(x, g) = \left|\left|\mu_x - \mu_g\right|\right|_2^2 + Tr\left(C_x + C_g - 2\left(C_x C_g\right)^{1/2}\right) \quad (20)$$

where μ_x and μ_g represent the mean values of the real and synthesized images, respectively, while C_x and C_g denote the covariance matrices of the real and synthesized images, respectively. FID is computed on the corresponding validation sets, comprising 302 images from GID-15, 462 images from ISPRS Potsdam, and 677 images from LoveDA Urban. For the main quantitative comparison in Table II, all reported results are obtained from a single training run with a fixed random seed. For segmentation-based metrics, we evaluated the mean intersection over union and pixel accuracy of pretrained semantic segmentation models on synthesized images. Let N represent the total number of categories, and let p_{ij} denote the probability that category i is predicted as j. The formulas for the mean intersection over union and pixel accuracy are presented in the following equations:

$$mIoU = \frac{1}{N}\sum_{i=1}^{N}\frac{p_{ii}}{\sum_{j=1}^{N} p_{ji} + \sum_{j=1}^{N} p_{ij} - p_{ii}} \quad (21)$$

$$Accuracy = \frac{\sum_{i=1}^{N} p_{ii}}{\sum_{i=1}^{N}\sum_{j=1}^{N} p_{ij}}. \quad (22)$$

We used DeepLabV3+ [63] as the segmentation model and trained it for 200 epochs.

4) Diversity Metrics: We used LPIPS [64] to assess global diversity. The LPIPS measures the weighted L1 distance. In order to evaluate diversity multidimensionally, we synthesize five images separately for each semantic mask and randomly sample five pairs of images for each mask to calculate the average LPIPS as the final result. Higher LPIPS scores indicate that the synthesized images have better global diversity.

5) Downstream Segmentation Improvement Metrics: To evaluate the performance improvement brought by synthetic images for downstream segmentation tasks, we first train a baseline model on the original training set and then compare its performance with that of the model trained on an augmented dataset that combines the original training set with synthetic images. This downstream experiment is distinct from the controllability metric above: here, the segmentation model is evaluated on real validation images with their real semantic labels, whereas the controllability metric compares predictions on synthesized images with the input masks used for generation. We use DeepLabV3+ [63] as the architecture, trained with the Adam [65] optimizer for the same epochs in both augmented and unaugmented settings. The performance metrics include mIoU, FWIoU, and accuracy, denoted as

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TABLE II
QUANTITATIVE COMPARISON ON THE GID-15, ISPRS, AND LOVEDA DATASETS

Method	GID-15				ISPRS				LoveDA			
	FID↓	mIoU↑	acc↑	LPIPS↑	FID↓	mIoU↑	acc↑	LPIPS↑	FID↓	mIoU↑	acc↑	LPIPS↑
Pix2PixHD [15]	206.114	49.930	74.217	-	185.589	58.571	80.010	-	153.950	45.580	61.650	-
SPADE [25]	142.579	57.044	82.673	-	144.340	67.168	86.382	-	119.101	61.830	72.430	-
SAFM [28]	139.791	64.318	86.601	0.151	156.077	65.902	86.510	0.349	120.057	66.360	74.330	0.437
CLADE [54]	145.655	59.591	81.069	0.187	153.128	68.413	87.537	0.335	132.694	63.230	73.120	0.510
INADE [24]	136.615	52.074	78.774	0.339	146.597	62.868	83.094	0.419	111.286	60.730	71.130	0.521
Santa [55]	157.844	-	-	-	173.354	-	-	-	226.144	-	-	-
CSEBGAN [7]	144.515	57.648	81.102	0.167	143.741	70.128	87.021	0.324	125.826	59.360	69.930	0.520
HySEGGAN [8]	132.283	**71.234**	88.022	0.189	134.195	77.403	91.980	0.317	112.620	**71.180**	**79.720**	0.413
FreestyleNet [22]	109.882	49.573	81.590	0.446	127.447	60.710	86.219	0.527	97.174	54.230	72.450	0.555
PLACE [23]	111.680	65.812	87.209	0.348	108.907	76.274	90.027	0.409	**90.717**	49.240	69.360	0.570
CRSDiff [46]	120.556	50.120	78.279	0.461	114.237	61.637	87.204	**0.570**	106.483	45.390	66.501	**0.615**
Ours	**106.879**	68.052	**88.921**	**0.487**	**105.824**	**77.705**	**92.220**	0.423	91.274	68.270	78.709	0.540

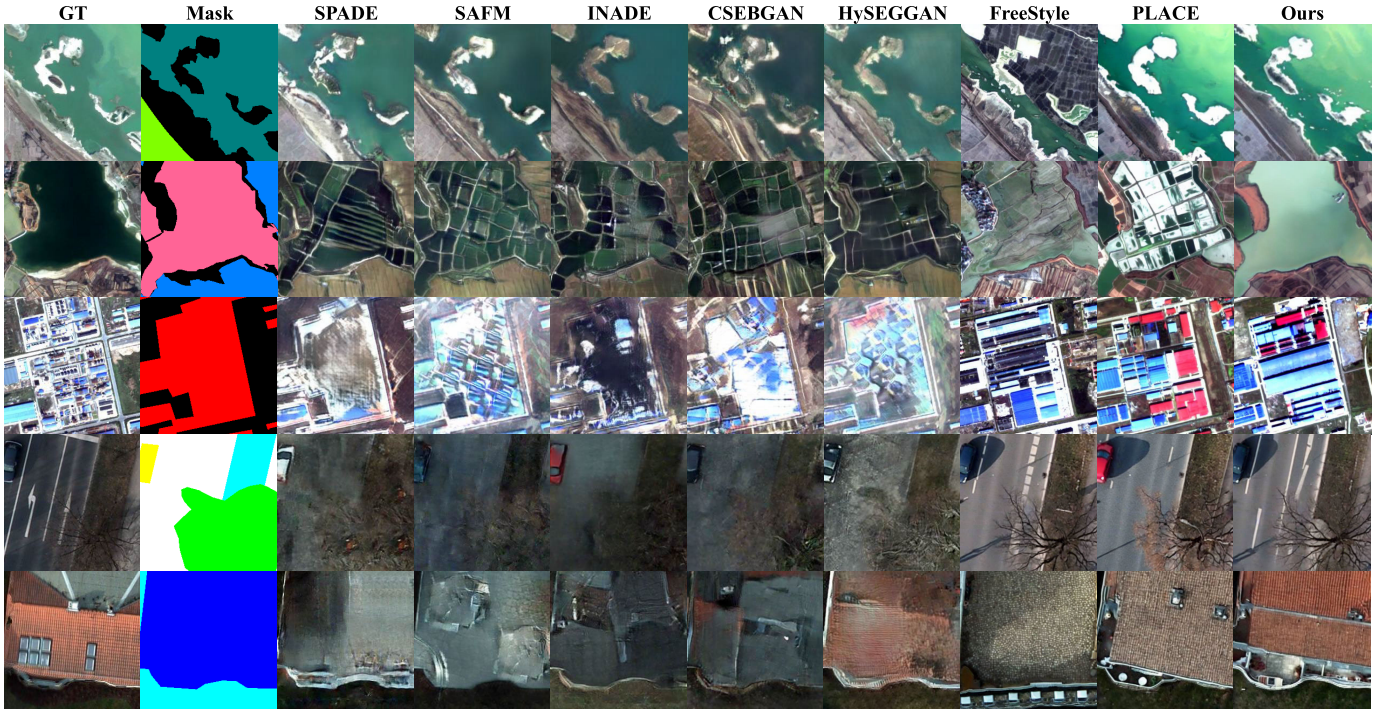


Fig. 3. Visual qualitative comparison on both GID-15 (*first–third* rows) and ISPRS (*fourth and fifth* rows) datasets.

d-mIoU, d-FWIoU, and d-acc. The formula for d-FWIoU is given in the following equation:

$$FWIoU = \sum_{i=1}^{N} w_i \cdot \frac{p_{ii}}{\sum_{j=1}^{N} p_{ji} + \sum_{j=1}^{N} p_{ij} - p_{ii}}. \quad (23)$$

The weight w_i for class i is calculated as the ratio of the number of pixels in the true label for that class to the total number of pixels.

B. Quantitative Comparison

We conducted a comprehensive quantitative evaluation, summarized in Table II. Our proposed method consistently achieves strong synthesis quality across the three datasets, obtaining FID scores of 106.879, 105.824, and 91.274 on GID-15, ISPRS Potsdam, and LoveDA Urban, respectively. On GID-15 and ISPRS Potsdam, CASE achieves the best FID scores, outperforming the second-best approaches by 3.003 and 3.083, respectively, while remaining competitive on LoveDA Urban. In terms of semantic consistency, CASE achieves strong mIoU and accuracy performance across all three datasets, demonstrating its ability to faithfully preserve the prescribed semantic layouts. Moreover, despite prioritizing synthesis quality and semantic controllability, our method maintains considerable conditional diversity, achieving competitive LPIPS scores across the three benchmarks. These results demonstrate that CASE provides a favorable balance among generation quality, semantic controllability, and diversity across diverse remote sensing scenarios. By leveraging a faithful CSM, our approach enhances the consistency of fine-grained details, achieving semantic controllability comparable to GAN-based methods while surpassing the fidelity of other

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TABLE III

QUANTITATIVE COMPARISON OF DOWNSTREAM SEGMENTATION IMPROVEMENT IS SHOWN ON BOTH GID-15 AND ISPRS DATASETS. SOURCE ONLY INDICATES NO SYNTHESIZED IMAGES WERE USED

Method	GID-15			ISPRS		
	d-mIoU↑	d-FWIoU↑	d-acc↑	d-mIoU↑	d-FWIoU↑	d-acc↑
source only	56.867 (+0.000)	68.250 (+0.000)	80.814 (+0.000)	67.102 (+0.000)	79.148 (+0.000)	87.107 (+0.000)
Pix2PixHD [15]	55.236 (-1.631)	68.644 (+0.394)	81.151 (+0.337)	66.950 (-0.152)	76.968 (-2.180)	87.350 (+0.243)
SPADE [25]	58.341 (+1.474)	69.421 (+1.171)	81.682 (+0.868)	67.960 (+0.858)	79.450 (+0.302)	87.848 (+0.741)
SAFM [28]	56.179 (-0.688)	68.903 (+0.653)	81.365 (+0.551)	67.113 (+0.011)	79.931 (+0.783)	87.670 (+0.563)
CLADE [54]	58.353 (+1.486)	69.522 (+1.272)	81.724 (+0.910)	68.250 (+1.148)	80.222 (+1.074)	88.020 (+0.913)
INADE [24]	57.681 (+0.814)	69.713 (+1.463)	81.873 (+1.059)	67.903 (+0.801)	80.248 (+1.100)	87.950 (+0.843)
Santa [55]	55.202 (-1.665)	68.263 (+0.013)	80.834 (+0.020)	68.105 (+1.003)	80.123 (+0.975)	88.009 (+0.902)
CSEBGAN [7]	58.349 (+1.482)	68.823 (+0.573)	81.374 (+0.560)	68.298 (+1.196)	80.236 (+1.088)	88.171 (+1.064)
HySEGGAN [8]	59.191 (+2.324)	**69.747 (+1.497)**	82.002 (+1.188)	68.480 (+1.378)	80.389 (+1.241)	88.240 (+1.133)
FreestyleNet [22]	56.043 (-0.824)	68.810 (+0.560)	81.297 (+0.483)	67.823 (+0.721)	80.007 (+0.859)	87.961 (+0.854)
PLACE [23]	59.619 (+2.752)	69.612 (+1.362)	81.978 (+1.164)	68.319 (+1.217)	80.504 (+1.356)	88.212 (+1.105)
Ours	**59.759 (+2.892)**	69.717 (+1.467)	**82.132 (+1.318)**	**69.331 (+2.229)**	**81.072 (+1.924)**	**88.919 (+1.812)**

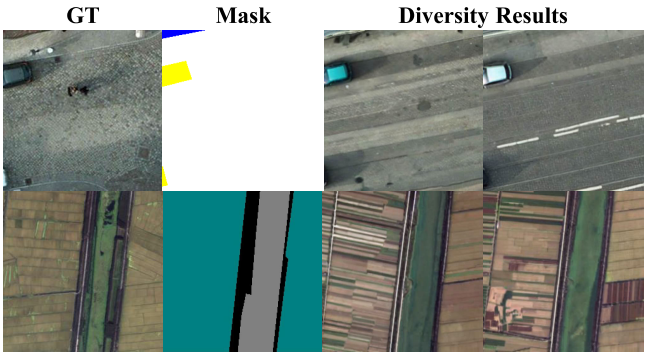


Fig. 4. Diversity images generated from the same semantic mask by CASE on both GID-15 and ISPRS datasets.

pretrained text-to-image models. Crucially, the integration of a novel FC function effectively mitigates semantic drift, a prevalent challenge in RSI synthesis. These results collectively indicate that our method not only achieves semantic fidelity competitive with leading models but also excels at preserving generation diversity.

C. Qualitative Comparison

As illustrated in Fig. 3 on the GID-15 (rows 1–3) and ISPRS (rows 4 and 5) datasets, our method generates images with superior realism and semantic fidelity. The proposed model demonstrates robust semantic control, accurately rendering the geometry of lakes (row 2) while maintaining structural coherence in dense buildings, mitigating the semantic discontinuity apparent in other methods (row 3). Furthermore, our approach excels in synthesizing high-fidelity textures and fine-grained details, evident in the sharp land-water boundaries (row 1), complex road elements (row 4), and consistent roofing styles (row 5). These results substantiate the effectiveness of our method for RSI synthesis, mitigating semantic drift and interclass confusion. To further examine conditional diversity, Fig. 4 presents the multiple outputs generated from the same semantic mask, illustrating variations in texture, color, and local appearance while preserving the prescribed semantic structure.

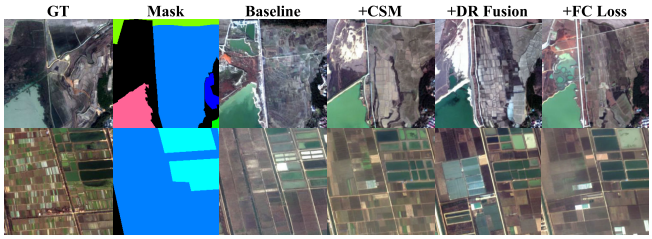


Fig. 5. Visual qualitative comparison of proposed modules on GID-15 dataset.

TABLE IV

INDIVIDUAL CONTRIBUTION OF THE PROPOSED MODULES ON GID-15 DATASET. (1) BASELINE + CSM. (2) BASELINE + DR FUSION. (3) BASELINE + FC LOSS

Method	CSM	DR Fusion	FC Loss	FID↓	mIoU↑
Baseline				112.512	62.712
(1)	✓			111.147	66.178
(2)		✓		109.752	62.225
(3)			✓	110.843	64.071
All	✓	✓	✓	**106.879**	**68.052**

D. Downstream Improvement of Synthesized Images

Table III reports a comparison of the performance of our method with other competitive methods in terms of mIoU, accuracy, and FWIoU. Our method achieved d-mIoU scores of 2.892% and 2.229% on the GID-15 and ISPRS Potsdam datasets. The d-FWIoU and accuracy scores ranked among the top two for both datasets. These gains are measurable, although the margins over strong baselines are moderate on GID-15. The quantitative results indicate that the synthesized images, as a data augmentation technique, effectively support the semantic segmentation task for remote sensing while ensuring semantic controllability and synthesis quality.

E. Ablation Study

1) Module Contribution: To validate the effectiveness of our approach, we conducted ablation studies on the GID-15 dataset. Our baseline employs a simple

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nearest-neighbor resized semantic map for token influence region determination, uses basic additive feature fusion, and is fine-tuned solely with denoising loss. The quantitative and qualitative results are presented in Table IV and Fig. 5, respectively. In Table IV, CSM denotes context-aware semantic map, DR Fusion refers to dynamic residual semantic fusion, and FC loss represents frequency constraint loss.

As shown in *baseline* and (1) in Table IV, the CSM significantly improved the mIoU score, increasing from 62.712% to 66.178%. Additionally, as shown in Fig. 5, using the CSM, our method not only generates images that closely align with the given layout (e.g., "pond" in the first row) but also enhances semantic coherence. In addition, as illustrated in *baseline* and (2) in Table IV, applying dynamic residual semantic fusion decreased the FID scores by 2.76. The results indicate that dynamic fusion improved the quality of the synthesized images. In Fig. 5, the results illustrate more realistic details (e.g., "urban" in the second row). Finally, as seen in the *baseline* and (3) of Table IV, the constraint loss leads to substantial improvements in synthesis quality and effectively mitigates semantic drift. The FID score decreased by 1.669, while the mIoU increased by 1.359%. As shown in Fig. 5, the FC loss also helps synthesize more detailed, realistic, and semantically consistent RSIs. Overall, the CSM primarily enhances semantic alignment, while DR Fusion mainly improves visual fidelity but may compromise semantic consistency when applied alone. The FC loss benefits both texture fidelity and semantic consistency, and the full model achieves the best performance by leveraging the complementarity of all components.

2) Impact of Frequency-Domain Transform Method: To validate the efficacy of the DWT as the foundational component of our frequency-domain constraint, we conducted a comparative analysis against another prevalent frequency-domain decomposition technique: the fast Fourier transform (FFT). In this ablation experiment, the DWT-based loss was substituted with an FFT-based counterpart, while all other components of the network architecture and hyperparameters were held constant.

1) *FFT-Based Loss:* We compute the 2-D FFT for the predicted clean latent $\hat{z}_0$ and the ground-truth clean latent z_0. The loss is formulated as the L_1 distance between the magnitudes of their Fourier spectra, while the phase component is not explicitly constrained. The phase component is intentionally disregarded due to its high sensitivity and the inherent difficulty in direct optimization, a common practice in frequency-domain loss formulations.
2) *DWT-Based Loss (Ours):* This is the proposed frequency-domain constraint, which leverages the bior4.4 wavelet basis.

The quantitative outcomes of this comparison are summarized in Table V. The results demonstrate that the DWT-based approach yields superior performance on both benchmark datasets, whereas the FFT-based constraint notably degrades quality even below the baseline. This degradation likely stems from the FFT loss's enforcement of a global frequency-domain magnitude constraint, which conflicts with the task's demand for precise local spatial fidelity. Conversely, this superiority is primarily attributable to the DWT's capacity to simultaneously capture spatial and frequency information, yielding a multiscale representation that effectively preserves the intricate textural details inherent in RSIs.

TABLE V
IMPACT OF FREQUENCY-DOMAIN TRANSFORM METHOD ON GID-15 AND ISPRS DATASETS

Method	GID-15		ISPRS	
	FID↓	mIoU↑	FID↓	mIoU↑
w/o FC Loss	109.598	65.091	108.896	76.511
FFT	118.991	64.748	117.095	76.471
Ours	**106.879**	**68.052**	**105.824**	**77.705**

TABLE VI
EFFECTS OF WAVELET BASIS SELECTION FOR IMAGE SYNTHESIS ON GID-15 DATASET

Method	FID↓	mIoU↑
w/o FC Loss	109.598	65.091
Haar	107.340	66.874
Coiflet	108.710	**68.122**
Symlets	108.584	66.914
Daubechies	107.092	67.207
Biorthogonal	**106.879**	68.052

3) Impact of Wavelet Basis: The efficacy of the frequency-domain constraint is intrinsically linked to the properties of the chosen wavelet basis, as different wavelets decompose the signal according to filters with varying characteristics. To determine the optimal basis for our RSI synthesis task, we conducted a series of experiments where we replaced the wavelet basis within our FC loss module, while keeping all other model parameters and training configurations constant. We evaluated five representative wavelet families: Haar, Coiflet (coif4), Symlets (sym4), Daubechies (db4), and Biorthogonal (Bior4.4). A baseline model trained without any wavelet loss was also included for reference. The quantitative results of our comparison are presented in Table VI. As shown, all wavelet-based variants outperform the baseline in both FID and mIoU, particularly in perceptual quality (FID) and semantic accuracy (mIoU), validating the value of frequency-domain guidance. Among the different bases, the biorthogonal wavelet (bior4.4) achieved the optimal FID score and the suboptimal mIoU performance.

4) Dynamic Residual Fusion Analysis: Our proposed fusion mechanism is governed by the dynamic fusion parameter λ_t, which is designed to adaptively balance the influence of the layout-centric contextual semantic map (A_{CSM}) and the original quality-centric attention map (A_{orig}) throughout the denoising trajectory. To validate the efficacy and behavior of this dynamic approach, we conducted two targeted analyses: 1) a visualization of the learned schedule for λ_t across diffusion timesteps and 2) a quantitative comparison against fixed fusion strategies.

We first sought to understand the strategy learned by ϕ_{scale}, which predicts λ_t from the timestep embedding t_{emb}. We

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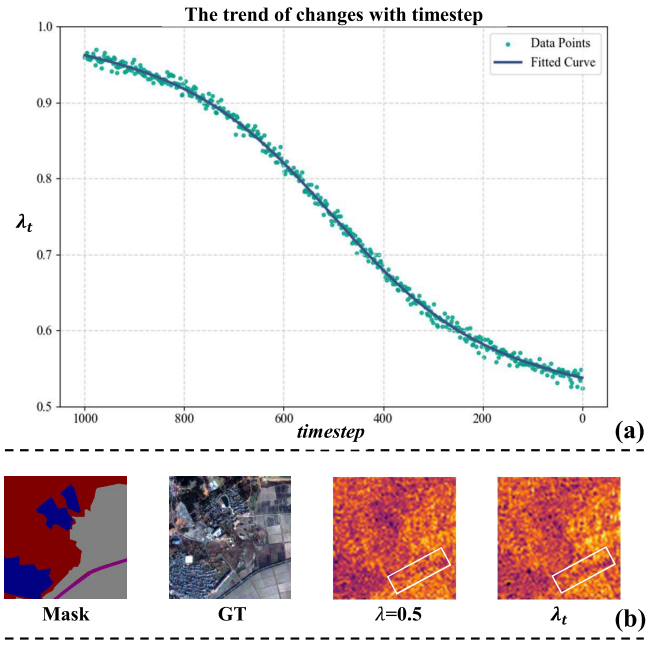


Fig. 6. Visualization of the dynamic fusion parameter. (a) Learned schedule for the dynamic fusion parameter λ_t as a function of the diffusion timestep t. (b) Feature map of the dynamic fusion parameter.

TABLE VII
IMPACT OF THE FUSION PARAMETER ON GID-15 DATASET

λ	0	0.25	0.5	0.75	1	λ_t
FID↓	111.097	110.104	108.120	108.469	111.825	**106.879**

plotted the output value of λ_t for each timestep $t \in [0, 1000]$. As shown in Fig. 6(a), the learned λ_t decreases smoothly along the denoising trajectory, starting from a relatively high value at large timesteps and gradually approaching a smaller value. In the initial stages of generation, the value of λ_t remains high. This indicates that the model heavily relies on our CSM A_{CSM} to establish the coarse layout and fundamental structure of the scene, which is critical when the signal-to-noise ratio is low. As the denoising process progresses into the intermediate and final stages, λ_t gradually decays toward a small value. This strategic shift allows the model to smoothly transition its focus, leveraging the rich, pretrained knowledge encapsulated in the original attention map A_{orig} to synthesize high-frequency textures and photorealistic details, thereby ensuring high generative quality. This learned behavior aligns with our initial intuition for the role of λ_t.

The effect of the fusion parameter is qualitatively validated in Fig. 6(b). The visualization compares internal feature maps produced using a fixed fusion weight ($\lambda = 0.5$) with those obtained by our dynamic fusion parameter λ_t. As the regions indicated in the boxes, the fixed strategy yields relatively ambiguous feature responses. In contrast, the dynamically learned λ_t exhibits substantially stronger representational power and aligns more faithfully with the ground truth, demonstrating superior ability to preserve fine-grained details.

TABLE VIII
COMPUTATIONAL EFFICIENCY OF CSM ON GID-15 DATASET

Method	Param. (M)	Time (Sec.)	FID↓	mIoU↑
Ours w/o CSM	859.5	13.3	107.761	64.324
Ours w/ CSM	863.2	15.9	106.879	68.052

TABLE IX
ROBUSTNESS TO DATA-LIMITED CONDITIONS ON GID-15 DATASET

δ	HySEGGAN			Ours		
	d-mIoU↑	d-FWIoU↑	d-acc↑	d-mIoU↑	d-FWIoU↑	d-acc↑
100%	+2.324	+1.497	+1.188	+2.892	+1.467	+1.318
75%	+2.273	+1.469	+1.276	+2.181	+1.451	+1.374
50%	+2.110	+1.507	+1.224	+2.357	+1.686	+1.435
25%	+1.871	+1.146	+1.039	+2.001	+1.311	+1.206
10%	+0.707	+0.015	-0.127	+1.531	+0.786	+0.641

5) Impact of Dynamic Fusion Parameter: To quantify the benefits of this adaptive schedule, we compared our dynamic λ_t approach against several fixed fusion strategies. We evaluated two baseline cases where only one attention map is used, A_{orig} only ($\lambda = 0$) and A_{CSM} only ($\lambda = 1$). Additionally, the models were trained with constant, manually set the values of $\lambda \in \{0.25, 0.5, 0.75\}$. The results are presented in Table VII.

6) Computational Efficiency: We further investigate the efficiency of our proposed CSM module. As summarized in Table VIII, we report the model parameters, inference time cost (sec.), and synthesis quality (FID) for configurations with and without the proposed CSM. Compared to the variant without CSM, the integration of layout-adaptive and context-aware features increases the parameter count by only 3.7M and the inference time by 2.6 s, while CSM reduces the FID score by 0.882 and improves 3.728 mIoU, demonstrating that CASE achieves an optimal tradeoff between structural consistency and inference efficiency.

7) Robustness to Limited Data: The practical application of generative models is often constrained by the lack of new samples and the difficulty of data annotation. To evaluate generalization and robustness, we conducted experiments under data-scarce conditions on the GID-15 dataset, with results presented in Table IX. Following the protocol in Section IV-A2, we constructed data-scarce subsets of the GID-15 dataset through random sampling from the training set, which substantially increases the difficulty of learning from limited data. We evaluate our method and the RSI synthesis method [8] on these limited subsets. To ensure the fairness of the experiment, we retrained all the generative models on each training subset and expanded each training subset with an equal amount of synthetic data using the trained generative models. Finally, we train each segmentation model on these augmented training subsets and achieve performance improvements by training with the augmented datasets. The results reveal that while both methods' augmentation benefits decrease with less data, our proposed method demonstrates significantly greater stability. The performance of the representative method [8] degenerates sharply in the

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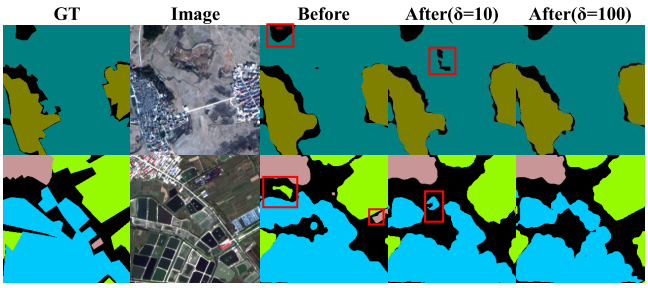


Fig. 7. Qualitative downstream segmentation comparisons under different data availability levels on the GID-15 dataset.

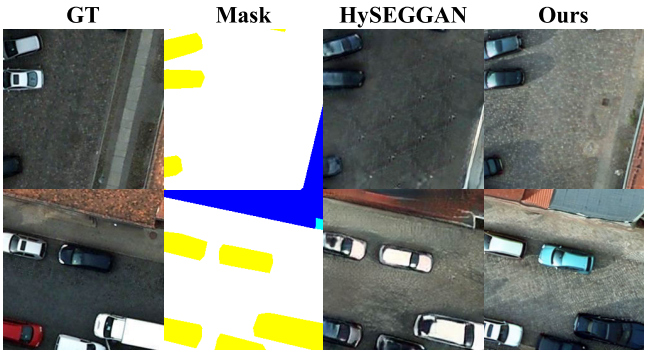


Fig. 8. Failure examples of the proposed method.

most constrained setting. Specifically, at the $\delta = 10\%$ level, HySEGGAN's augmentation benefit becomes minimal and even degrades downstream segmentation accuracy, indicating a failure to capture a useful data distribution. In stark contrast, our method sustains a robust improvement even with minimal data, delivering a +1.531 d-mIoU gain. These results suggest that CASE provides more stable augmentation benefits under the evaluated data-limited protocol.

To complement these quantitative results, Fig. 7 provides the qualitative downstream segmentation comparisons. For representative scenes, the figure presents the input image, ground-truth semantic map, and predictions obtained with source-only training and CASE-augmented training. Following Section IV-E7, we select $\delta = 10\%$ and $\delta = 100\%$ as representative settings, corresponding to the most data-scarce and full-data scenarios, respectively.

8) Failure Example and Analysis: Despite achieving advanced performance across various scenarios, our model, such as all deep generative models, possesses inherent limitations. This section presents a detailed analysis of typical failure cases. As depicted in Fig. 8, while superior to the representative method [8], some generated instances still suffer from minor structural artifacts and a loss of fine-grained detail, appearing smoother or slightly malformed. For example, the generated cars in the first row exhibit significant distortion, while those in the second row lack structural integrity. These failures primarily stem from the limitations of diffusion models operating in a low-resolution latent space, which impedes their ability to precisely model fine-grained details and small objects. Therefore, the generation of multi-instance RSIs remains a critical challenge for future work, particularly regarding fine-grained feature alignment and consistency modeling across different scales and instances. Moreover, the current evaluation is limited to the considered optical datasets and experimental protocols, while cross-sensor, cross-resolution, multimodal, and open-world generalization remain important and challenging directions for future research.

V. Conclusion

In this article, we presented CASE, a novel framework that advances RSI synthesis by enhancing the semantic controllability of pretrained diffusion models. Our approach resolves the critical tension between layout fidelity and generative quality through a CSM and a dynamic fusion module. Furthermore, the proposed FC loss provides empirical improvements in multiscale structural and textural consistency. Extensive evaluations depict that CASE delivers favorable synthesis quality and competitive semantic controllability on three remote sensing benchmarks. The method also improves downstream segmentation performance when used for data augmentation, although some gains over strong baselines are moderate. These results indicate that frequency-based and layout-guided diffusion fine-tuning is a promising direction for RSI synthesis.

Concurrently, our investigation highlights the intricate challenge of high-fidelity multi-instance synthesis of RSIs. We observe that preserving fine-grained structural integrity and mitigating distortion, particularly within dense clusters of foreground objects, persists as a formidable open problem for the field. In future work, we aim to focus on instance-aware layout modeling and attention mechanisms for fine-grained and multi-instance RSI synthesis.

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